

كلية الطب – جامعة العميد

الحملة التوعوية حول الحصول على

الاعتمادية

الرؤية

التميز الفعال من اجل رفد المجتمع بأطباء اكفاء في الخدمة الطبية والبحث العلمي

الرسالة

توفير بيئة جامعية تعمل بمنهج تعليمي مواكب للتطور
اطباء ذوي كفاءة ومستوى علمي قادرين على تقديم الخدمة المتميزة



الاهداف

- اعداد اطباء بكفاءة عالية
- التأكيد على مبادئ الاخلاق الطبية
- نشر مفاهيم العمل كفريق واحد
- العمل على حل المشكلات الصحية
- تطبيق مخرجات البحوث المتقدمة
- تعزيز عملية البحث العلمي
- تبني مفاهيم التعلم المهني المستمر



University of Al-Ameed
College of Medicine
Department of Dharmacology



Introduction- Drug distribution

Assist. Professor Dr. Kamal Al-Yassiry

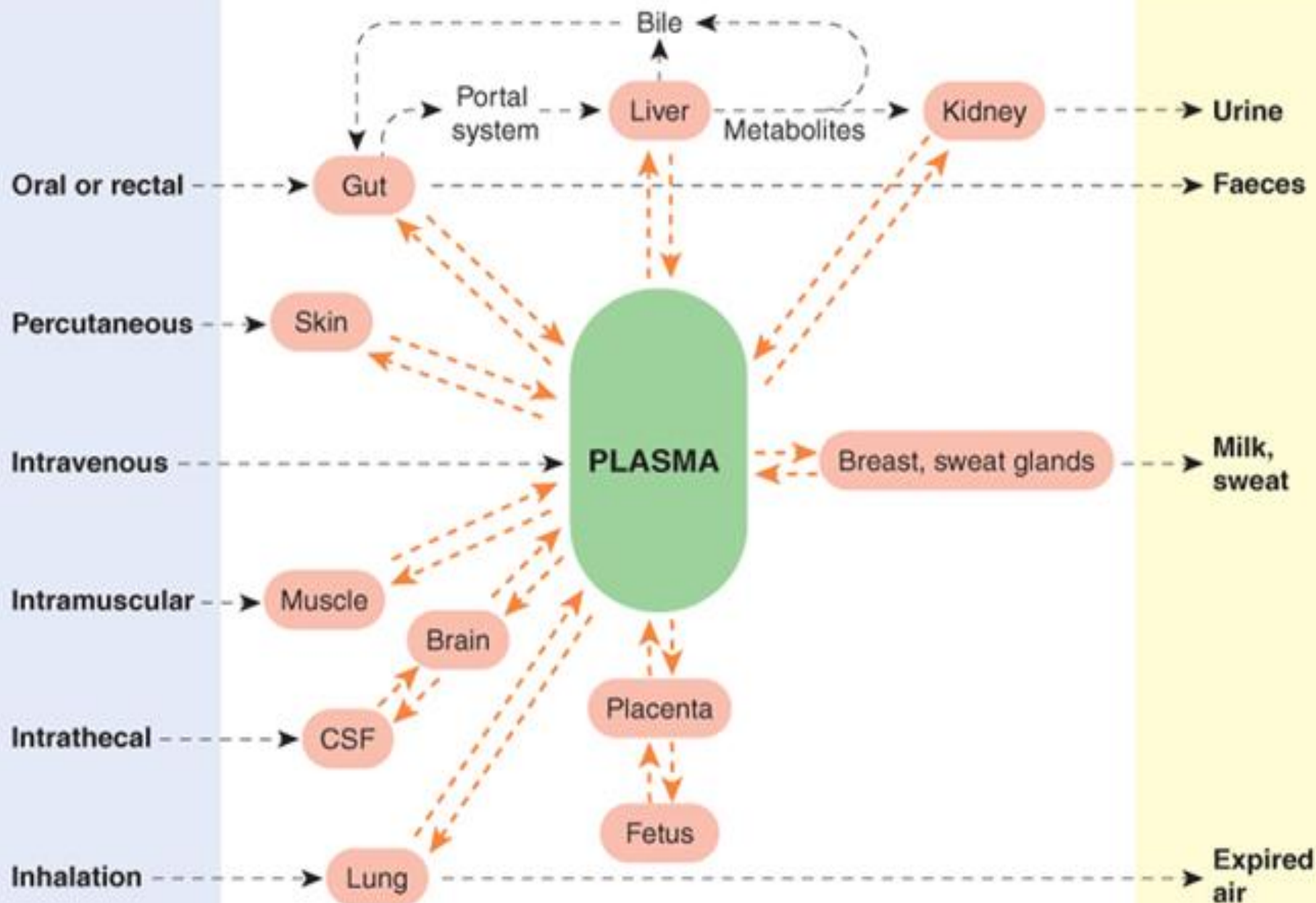
Pharmacologist

Drug Distribution

- Drug distribution is the process by which a drug reversibly leaves the bloodstream and enters the interstitium (extra cellular fluid) and/or the cells.

The delivery of a drug from the plasma to the interstitium primarily depends on:

- ❖ *Blood flow*
- ❖ *Capillary permeability*
- ❖ *The degree of drug binding to plasma proteins*
- ❖ *The relative hydrophobicity of the drug.*

Administration**Absorption and distribution****Elimination**

A. Blood flow

- The rate of blood flow to tissues varies widely, as a result, unequal distribution occurs.
- Blood flow to the brain, liver, and kidney is greater than that to the skeletal muscles, most viscera, skin, and fat.
 - e.g. The high blood flow, together with the high lipid solubility of *thiopental*, permit it to rapidly move into the CNS and produce anesthesia.

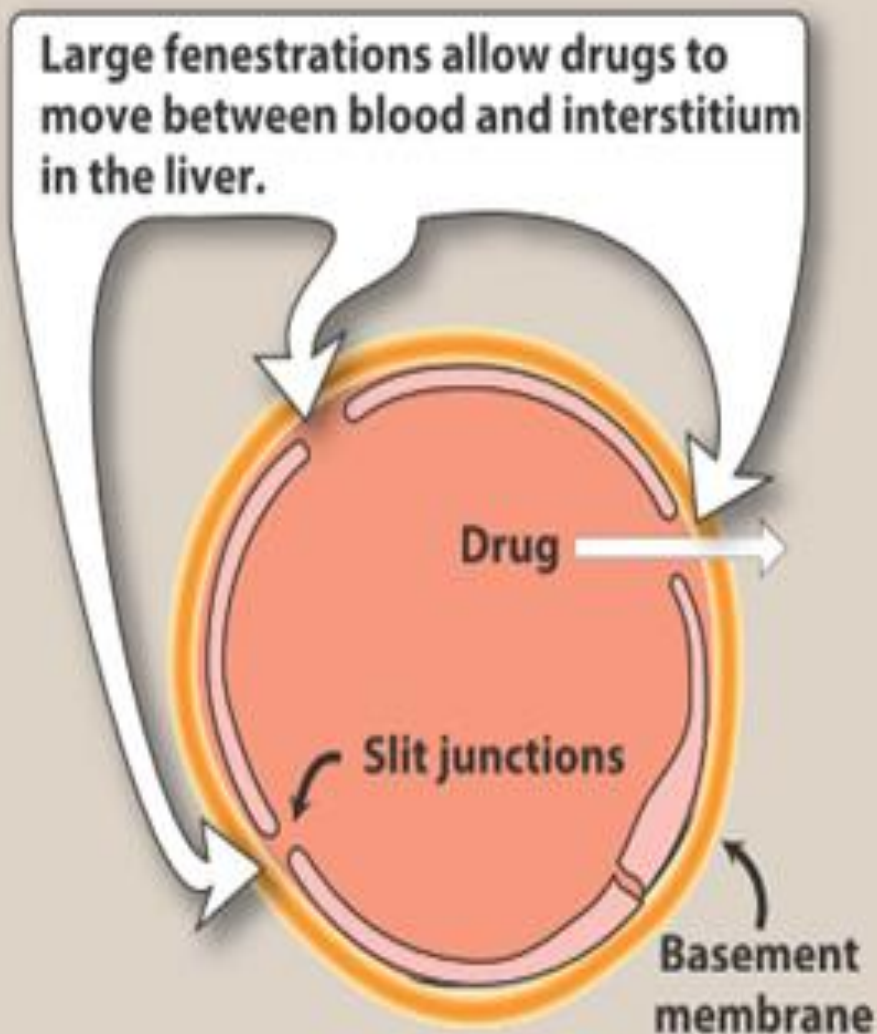
B-Capillary permeability

- ✚ It determined by capillary structure and by the chemical nature of the drug.
- ✚ To enter the brain, drugs must pass through the endothelial cells of the capillaries of the CNS or be actively transported like levodopa. Lipid-soluble drugs readily penetrate into the CNS because they can dissolve in the membrane of the endothelial cells. Ionized or polar drugs generally fail to enter the CNS because they are unable to pass through the endothelial cells of the CNS.

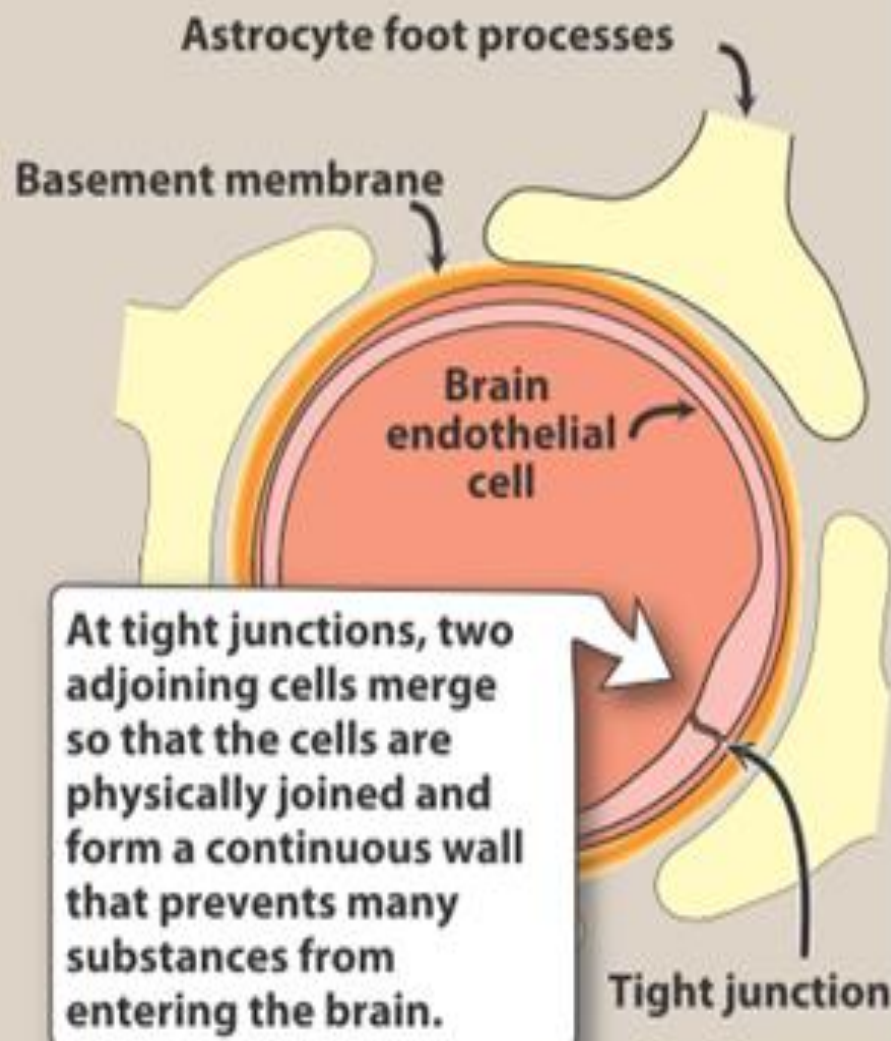
A

Structure of endothelial cells in the liver

Large fenestrations allow drugs to move between blood and interstitium in the liver.

**B**

Structure of a brain capillary



C-The degree of binding of the drug to plasma and tissue proteins:

- ✚ Reversible binding to plasma proteins sequesters drugs in a non-diffusible form and slows their transfer out of the vascular compartment.
- ✚ Plasma **albumin** is the major drug-binding protein and may act as a drug reservoir.
- ✚ Bound drugs are pharmacologically inactive; only the free, unbound drug can act on target sites in the tissues.
- ✚ Albumin has the strongest affinities for hydrophobic drugs, while hydrophilic drugs do not bind to albumin.

Competition for binding between drugs:

- ❑ When two drugs are given, each with high affinity for albumin, they compete for the available binding sites, and can result in drug displacement.
- ❑ Example on the Clinical importance of drug displacement: (Warfarin is highly bound to albumin, and only a small fraction is free. If a sulfonamide is administered, it displaces warfarin from albumin, leading to a rapid increase in the concentration of free warfarin in plasma; the increase in warfarin concentration may lead to increased therapeutic effects, as well as increased toxic effects, such as bleeding).

Plasma protein binding for some drugs:

Drug	p.p. binding	Drug	p.p. binding
Warfarin	99 %	Morphine	35 %
Diazepam	98 %	Digoxin	25 %
Amitriptyline	95 %	Amoxycillin	18 %
Trimethoprim	70 %	Ethosuximide	0.0 %

The relative hydrophobicity of the drug:

- ✚ Hydrophobic drugs readily move across most biologic membranes.

Volume of distribution:

- ❖ Apparent Volume of distribution (V_d) relates the amount of drug in the body to the concentration of drug (C_0) in blood or plasma:
- ❖ V_d is a useful pharmacokinetic parameter *for calculating a drug's loading dose*

$$V_d = \frac{\text{Amount of drug in the body}}{C_0}$$

Distribution into the water compartments in the body:

1. Plasma compartment (6% of the body weight): large molecular weight drugs will be trapped here (cannot pass through endothelium) so it will bind extensively to protein like heparin.

2. Extracellular fluid (20% of the body weight): hydrophilic, low molecular weight drugs will distribute here (it is the sum of the plasma water and the interstitial fluid) like aminoglycosides.

3. Total body water (60% of body weight): hydrophobic, low molecular weight drugs will distribute here like Ethanol.

Drug	V_d (L / 70kg)	V_d (L / kg)
Heparin	5	0.07
Aspirin	11	0.15
Amoxycillin	28	0.4
Atenolol	77	1.1
Diazepam	140	2.0
Digoxin	420	6.0
Nortriptyline	1000	14.2
Chloroquine	13000	185.7

Any factor that increases V_d can lead to an increase in the $t_{1/2}$ and can extend the duration of action of some drugs.